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Design Principles of Hand Gesture Interfaces for Microinteractions

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ABSTRACT

We experience a drastic increase in post-desktop input devices and interaction techniques but still lack in specific and applicable design principles for these systems. This paper presents a set of design principles for hand gesture based microinteractions. The main concepts from related work are fused together in order to build a clear structure that allows heuristic evaluation. Moreover, a visualization of a *gesture phrase* helps understanding the relationship of crucial concepts such as feedforward/feedback and a gesture's tension. The applicability of the proposed design principles is then exemplarily shown by the development of a truly ubiquitous interactive system for hand gesture based microinteractions.

Author Keywords

Design principles; microinteractions; gesture interfaces; ubiquitous computing; smart environments; feedforward; self-revealing.

General Terms

Design; Theory.

INTRODUCTION

Over the last decade we have experienced a drastic increase in production of post-desktop multi-purpose devices like smartphones and tablets. Although their general GUI nature allows to perform tasks of different complexity levels – from checking the time to writing an email – users face some sort of efficiency–availability trade-off. For example, users might prefer a desktop setup with a standard keyboard/mouse input and a large display for performing complex long-term tasks. At the same time, for years there has been a big variety of special-purpose devices, such as wearable watches or mp3-players that exhibit UI-simplicity and appropriate physical properties, e.g., small number of haptic-discriminable knobs. Such physical features allow the user eyes-free control of an mp3-player without taking

it out of the pocket, thus decreasing the time of interaction. Some of the issues concerning the placement of devices on a user's body in mobile usage scenarios were investigated in Ashbrook's dissertation “Enabling mobile microinteractions” [1]. He found out that the pocket access time of a mobile phone lasts on average 4.6 seconds. Furthermore, he defined microinteractions as “interactions with a device that take less than four seconds to initiate and complete”. These are non-main task interactions that are performed on the go without distraction from the main task, e.g., controlling the music while driving a car or riding a snowboard.

Over the last few years directed research has been conducted in the theory of microinteractions [2, 59, 62] as well as in the area of interfaces enabling *mobile* microinteractions on handheld devices [6, 21, 60, 61] and wearables [20, 22, 37, 43, 50, 63]. Some *stationary* systems integrated into smart environments support the concept of microinteractions, but they cannot be considered always-available, as availability depends on the user's location in space. Such integrated interfaces should rather offer affordances in the environment that augment everyday surfaces and objects with new input techniques [13, 20, 44, 46, 56].

It should also be noted that ubiquitous computing principles proposed in the 1980s and 90s found application in the beginning of 21st century [23], as sufficient mass technology become available. At the moment, we also observe a boom in new commercial always-available wearable interfaces, e.g., Google Glass¹, Pebble², MYO³. All of these devices support microinteractions to some extent but have different characteristics in terms of input and output modalities.

This paper focuses on hand gesture interfaces for microinteractions. Hand gesture interfaces have an important property that makes them preferable over speech interfaces for microinteractions in most usage contexts. In

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¹ See <http://google.com/glass/start/>

² See <http://getpebble.com/>

³ See <http://getmyo.com/>

order to explain this, we should refer to the paper of Shneiderman [48] in which he describes that in contrast to speech processing tasks, tasks of pointing and clicking use other brain resources than problem solving and recall. Therefore, while performing a main task the use of speech for a non-main task would involve more cognitive load than gesture and as a consequence would distract a user from the main task. In addition, the limitation to hand-gesture input makes the design space less general but brings appropriate simplicity which makes the design space more applicable for practitioners.

The contribution of this paper is twofold. The first part contributes to the exploration of design principles of hand gesture interfaces for microinteractions. By reviewing and summarizing the main concepts from literature and existing systems related to hand gesture interfaces we derived a structure which is a helpful design instrument for researchers and practitioners.

The second contribution of this paper is an exemplary application of the proposed design principles. We designed an unobtrusive ubiquitous system for hand gesture microinteractions. It is based on data collected from depth-cameras. By using a self-revealing menu projected on an “every-day” surface, a user can learn appropriate on-surface gestures while interacting with the menu. In addition to common home automation applications (e.g., to control lights or doors) the concept of microinteractions is used to provide the inhabitant with often used and helpful functions, such as audio reminders or control of music.

Despite the fact that we focused on the unobtrusive interaction required by the smart environment, the proposed guidelines can also be applied to gesture-based microinteractions for handheld as well as wearable devices.

RELATED WORK

Although the main contribution of the paper lies in the exploration of the design principles including but not limited to unobtrusive interfaces for microinteractions, the smart environment context of the exemplarily developed system determines the related work below. We focus mostly on techniques that allow unobtrusive augmentation of everyday surfaces, without changing appearance or material characteristics of objects. In addition, implementing a pie menu in the system and thereby using visual feedback increases the importance of ubiquitous displays.

Ubiquitous Displays

The idea to augment the existing environment with computer-aided imagery is not new. Already in the mid-1970s, Myron Krueger created a so-called *VideoPlace* [30] in which he combined multiple projectors and cameras to provide the user with new ways of interaction. Silhouettes of users, extracted from a video stream, were projected on walls, thereby allowing this silhouette to be an interaction medium with virtual objects (in the same projection).

The use of multiple projections to augment the existing environment in a ubiquitous way was well described in a visionary “*office of the future*” by Raskar et al. [42]. They suggested the use of *imperceptible structured light* to extract per-pixel depth and reflectance information for all visible surfaces, thereby allowing automatic adjustment of projections for corresponding surfaces. Another technique was proposed by Tokuda et al. [51] in which they used a special range sensor (time-of-flight) to dynamically model the shape of the whole room.

Pinhanez et al. [40] introduced the *Everywhere Displays Projector*, a device that incorporates a rotating mirror to project onto different surfaces. The use of an additional camera allows to compensate for perspective distortion and enables touch interaction with the surface.

It can be seen that the concept of ubiquitous displays consists of two main parts: projection and dynamic 3D modeling of the space. While projection technologies have gradually evolved for decades in terms of resolution, size, and energy consumption, 3D modeling technology experienced a drastic improvement in the last five years with new and cheap depth-sensing technology based on the recognition of IR-pattern deformations.

Ubiquitous Device-Free Input Techniques

In 2002, the first implementation of fully unobtrusive interaction for domestic environments took place: *LightWidgets* by Fails and Olsen [13]. The system with two cameras enabled to recognize fingers touching a surface. Every visible surface could be used as an interactive widget controlled with palm movements, e.g., the edge of a surface could be used as a slider to dim the light.

Another way of finding the palm position on an everyday surface was proposed by Schmidt et al. [46]. Their method utilizes embedded load sensors to calculate the load distribution over the surface and is not as unobtrusive as other techniques mentioned here.

Appearance of new depth-sensing technology based on the recognition of an IR-light pattern deformation also allows an accurate detection of a touch without any changes to the construction of a surface [56].

Microphones attached to a surface enable acoustic-based input that utilizes the unique sound of dragging a fingernail over a textured material [20]. This method allows to recognize a set of different gestures performed by the dragged fingernail. In addition, the newly proposed *Swept Frequency Capacitive Sensing* technique enables recognition of a variety of touch gestures with conductive materials [44].

Ubiquitous Surface Interaction

The paragraphs above considered ubiquitous input and output methods, but only the combination makes the interactive system useable. Broad expansion can be observed especially in the union of the depth-camera and

projector. In 2009, small context-aware menus to control everyday objects based on sticky notes were proposed [36]. For the accurate recognition of a note and the correct projection of a menu, IR markers were placed on them. Another project that utilizes the set of depth-cameras and projectors for un-instrumented surfaces is the *LightSpace* [57]. In addition, there are already toolkits available for rapid development of ubiquitous interfaces that use the same combination of input/output methods [17].

Similar techniques are also used for wearable interfaces. One example of such a system that is based on a pico-projector and gesture-recognition of marked fingers with a normal camera is *WUW* [37]. The *OmniTouch* interface [19] is very similar yet differs in the usage of a small depth-camera that enables touch detection. Both of these interfaces are not limited to un-instrumented surface, such as tables or walls, but also support interaction with small dynamic surfaces (e.g., forearm, palm, or a piece of paper).

DESIGN PRINCIPLES

Over the course of time multiple HCI design principles have been created and discussed [7, 39, 47]. On the one hand these heuristics can be generally helpful in any HCI project. On the other hand they lack specialization while being applied to a specific project: in our case, a hand gesture interface for microinteractions. Design principles as well as models or design spaces have similar issues concerning a completeness—applicability trade-off. Our resulting trade-offs are based on a small number of design principles, each of which has a dense related knowledge. Thus, multiple related taxonomies and important concepts are considered only in relation to a specific design principle. Although we have tried to provide a structure in terms of logical separation and proper order of appearance, a few design principles overlap to some extent. Therefore, some concepts are mentioned multiple times.

Overview of Design Principles

The authors of *Charade* [5] discussed various concepts and problems that arose during the development of one of the first gesture interfaces. Some of these issues are considered and broadened in the current paper, such as *incremental actions*, *tension* and *fatigue*. Apart from gesture interface principles we also consider those for smart environments. The authors of a publication about a table that can be used as a pointing device [46] proposed various principles concerning a ubiquitous augmentation of an everyday surface, which are also integrated in our principles. Therefore, there are two main parts in the proposed design principles: one part is related to hand gesture interfaces for microinteractions and the second to smart environments generally.

Moreover, the area of gesture interfaces is also quite extensive. Our principles were designed to fit one kind of gesture interfaces: interfaces that deal with microinteractions. Although, systems such as *Charade* [5], “*put-that-there*” [8], and *g-stalt* [65] represent another class

of gesture interfaces in which the user performs gestures for the main task, some interaction parts of those systems can be considered as suitable for microinteractions. In any case, the context of use for these systems is a prolonged interaction with a computer. The flow of the interaction in these systems for a main task and for microinteractions is varied and mainly differs in the activation phase and segmentation of gestures. Therefore, the design principles for microinteractions described later on are less suitable for other forms of gesture interfaces.

Before we start with the discussion of the principles we should first refer to Buxton's idea of phrasing in human-computer dialogue that was also mentioned in *Charade*. First of all, Buxton [10] borrows two concepts from the music, namely *tension* and *closure*: “during a phrase there is a state of tension associated with heightened attention. This is delimited by periods of relaxation that close the thought and state implicitly that another phrase can be introduced by either party in the dialogue. [...] In manual input, I will want tension in imply muscular tension”. We use both concepts to describe the interaction flow in design principles. For instance, recognition techniques for hand gesture segmentation can be based on *tension* [18].

In addition, Buxton discusses the idea of chunks concerning words and phrases. He compares words with atomic tasks in HCI and argues that humans speak with sentences and not with words. As a consequence, he affirms: “my thesis is, if you can say it in words in a single phrase, you should be able to express it to the computer in a single gesture. This binding of concepts and gestures thereby becomes the means of articulating the unit tasks of an application”.

As a matter of fact, the notion of *gesture phrase* was broadly used by Kendon [28] in his research on co-speech gestures in human—human interaction. The metaphor of a phrase used by Buxton differs from the Kendon's one. While Buxton considers the phrase as a unit of a few single gestures, Kendon divides a single co-speech gesture into five phases and defines it as a gesture phrase.

Later on, we use the Buxton's consideration of a phrase with few adjustments in the idea of chunking. Gesture is an extensive notion which basically describes any body movement or its combination. Under a single gesture Buxton understands one chunk, a gesture without any (conscious) discontinuity. Though he takes the metaphor of a phrase from human—human interaction, there are limitations of the extent for comparison of human—human and human—computer interactions. In human speech, feedback does not play such a crucial role as in the dialog with a computer. But this is exactly the feedback that creates discontinuity in the interaction because the user consciously pays attention and reacts if the feedback is positive. Thus, we define a *gesture phrase* not as one chunk, but as a sequence of chunks (single gestures) that have one common, logically closed intent.

For the simplicity of the interaction the number of single gestures in one phrase should be kept as small as possible. In the exemplary system gesture phrases contain basically only two single gestures: one activation gesture and one gesture that is learned during the interaction with the pie menu.

As already mentioned, the proposed design principles can be logically divided into two parts. The first part deals with the specifics of hand gesture interaction for microinteractions:

1. *The Gesture Phrase* unites a sequence of single gestures and the system's reactions into one segment and defines one command, see figure 1:
 - a. *Activation*: an unconventional and arduous activation gesture ensures that the system does not start accidentally.
 - b. *Feedforward and Feedback*: provide continuous feedforward and feedback (e.g., system attention, from Charade).
 - c. *Incremental Actions*: support (additional) different single gestures within the same gesture phrase (e.g., next/previous song).
 - d. *Closure*: the last phase helps to identify the end of the gesture phrase.
2. *Self-revealing*: facilitates the interaction for a novice as well as the gesture-learnability for an expert.
3. *Recovering from errors*: provides the undo-functionality with a simple and solid gesture.
4. *Fatigue*: gesture interaction involves more muscular tension than standard interfaces, therefore gesture commands must be concise.

The second part contains three points concerning ubiquitous reality based smart environments that are mostly borrowed from [46]:

1. *Unobtrusiveness*: preserving the original functionality and the genuine appearance of the environment.
2. *Reliability*: solid and “calm” implementation.
3. *Sustainability*: energy efficient realization.

We concentrate more on the first part of the design principles. Special attention will be given to the description of the gesture phrase and its components that are visualized with a kind of (intuitive) infographic, see in figure 1.

Gesture Phrase

Human life is a perpetual movement of the whole body and its parts. In other words, we gesture all the time. In

Charade, a sensing system must identify and interpret certain gestures as a sequence of commands and “must have a way of segmenting the continuous stream of captured motion into discrete *lexical* entities”. The authors of Charade used the *segmentation of hand gestures* notion to describe this issue. We substitute this notion with the *gesture phrase*. This already implies that the system separates continuous stream and unites several discrete gestures into one segment.

Also, we consider the gesture phrase as a minimal set of chunks (single gestures) united by a common intent of a microinteraction.

Activation

One of the most critical parts of a gesture interface is the way how the user shows his intent to communicate with the system. This problem was named as *immersion syndrome* in Charade [5] and “Address: Directing communication to a system” in a paper by Bellotti et al. [7].

As described by Buxton, there is a first “state of tension associated with heightened attention” in the gesture phrase. Therefore, the first activation gesture is associated with maximum muscular tension and unconventional hand/finger movements. At the same time, this gesture must be natural and intuitive. Selection of an appropriate activation gesture could be guided by Cadoz's taxonomy of hand movements [11] and can help a system designer to ponder between the rarity of the gesture and its naturalness. Cadoz divided all hand movements into three groups:

1. *Epistemic*: used for perception, performed to explore the environment, for example checking the presence of the wallet in the back pocket.
2. *Semiotic*: used for human-human communication. Such gesticulations provide the discourse with additional information.
3. *Ergotic*: hand movements used to manipulate the environment. For example, gestures used to interact with touchscreen are ergotic.

If an environment is intended for the human-human communication, the use of semiotic gestures is not the best solution. Epistemic hand movements that mostly consist of (haptic) contact are not as expressive as ergotic gestures. Therefore, we propose to use ergotic hand movements for the activation gesture.

In order to overcome the *immersion syndrome* problem the authors of Charade proposed the use of an *active zone* which must be used for a hand to trigger the system for the gesture input. In the case of the presentation in Charade the active zone is a whiteboard. Appropriate combination of *active zone* and activation gesture could also be a good practice. However, a more general solution to the *immersion syndrome* utilizes only an activation gesture.

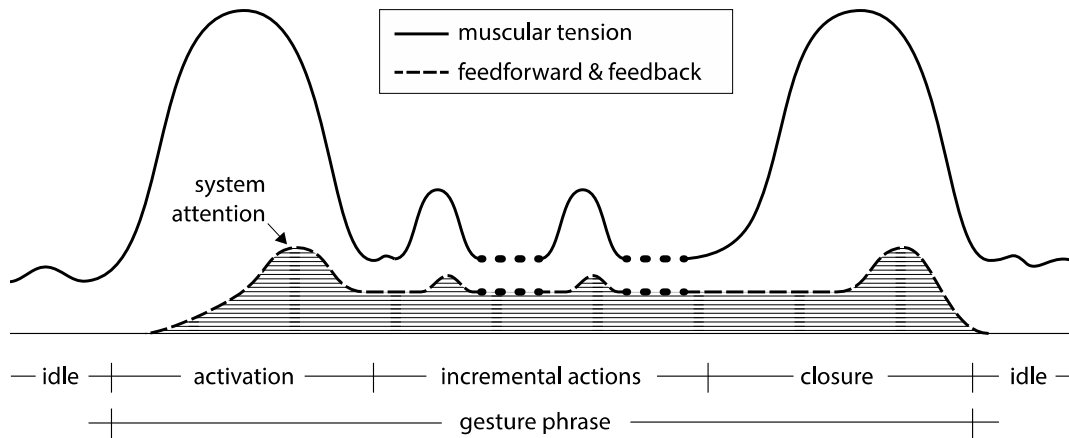


Figure 1: Schematic representation of a gesture phrase along with feedforward/feedback continuity

Moreover, so-called dwell-based interactions [53], “point and wait” or “point and shake” strategies [45] can be counted as a sort of active zone. They are widely employed in gaze and gesture interfaces: dwell or a combination of dwell and shake helped to overcome the *Midas-Touch* problem [27]. This interaction technique is mostly used to select virtual objects but it can also be used to control (select) physical objects in the environment. These techniques can be utilized for a time based activation. On one hand, this interaction is based on a natural pointing gesture. On the other hand, this time-based interaction does not allow a user to control the interaction time that would be appropriate for a novice user but not for an experienced one. In addition, precision issues of the pointing interaction should also be considered [55], along with the environment setup. Under environment setup we understand the position of objects with respect to the user's point of observation. Objects can be concealed by another one, making the selection by simple pointing impossible. At the same time, an additional “skip” interaction allows a user to switch between objects lying on one line-of-sight.

Feedforward and Feedback

The second important principle corresponds to the concept of feedback and feedforward [52]. When the user starts to communicate with the system and performs an activation gesture, the system must respond and indicate that it “listens” to the user, referred to as *system attention* by Baudel et al. [5]. This is analogous to human-human communication in which attention is shown through gaze and eye-contact. To enrich this phase all system output modalities should be used, such as visual, auditory, or haptic feedback. These types of feedback can vary: overall (sounds from main speakers) or in-place (door lock buzzing). In addition, this feedback combination should be harmonious and should support a metaphor of the activation gesture. We believe that the feedback concept should be as

continuous as possible [4] and suggest implementing *feedforward*.

At the beginning of the activation gesture there is some uncertainty, which should also be visualized by the system, to accompany the user with the calm feedforward/feedback during the entire activation gesture. Figure 2 shows one peak of a gesture tension along with continuous system reaction. We consider feedforward as a system reaction before the *discrete execution* of a command occurs. This peak is a pattern for each of the peaks in the gesture phrase. Time difference between the start of the activation gesture and the start of the first feedforward is shown with the *feedforward threshold*. Meanwhile, if the feedforward threshold is not enough, frequent appearing of feedforward might be quite disturbing for the user. Therefore, system designers have to trade-off between continuity and calmness of feedforward/feedback.

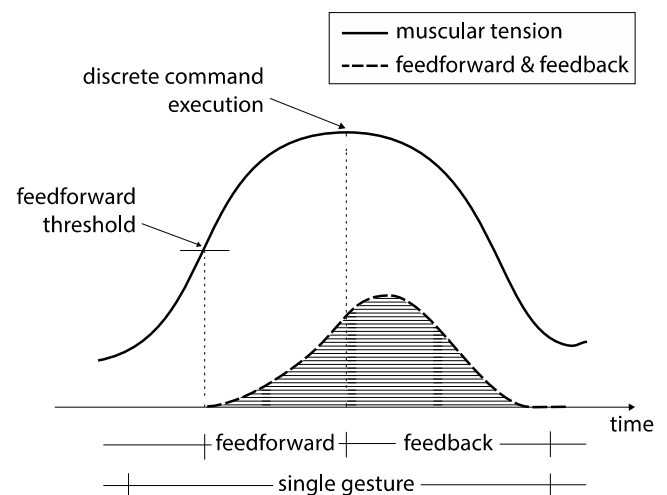


Figure 2: Schematic representation of a single gesture along with feedforward/feedback continuity.

Incremental Actions

Once the activation gesture with the maximum tension is performed, a user executes the main gesture, associated with a specific command. A microinteraction gesture as a shortcut (e.g., toggle-close of a window) contains only one command. In contrast, multiple microinteraction tasks (e.g., going through a list of songs) require incremented commands which are bound by common logic and closed intent. Thereby, the middle part of the gesture phrase can contain a single (shortcut) as well as multiple *incremental* gestures separated by feedback/feedforward periods.

Closure

Buxton borrowed the notion of closure from music's "period of relaxation". There might be multiple solutions for *closure*. Similar to releasing the mouse-button while performing mouse-gestures, leaving the active zone signifies the end of the hand gesture input. An alternative that we propose as more general relates to a special closure gesture that has similar characteristics as the activation one.

Self-Revealing

One of its main characteristics of a gesture interface is the possibility to teach novices to use gestures while interacting with the system. PIXIE [58] started the development of different self-revealing crossing-based radial menus: pie menus [12], marking menus [20, 32, 33], simple mark menus [65], zone and polygon menus [64], flower menus [3], multitouch marking menus [35], or marking menus for mid-air gestures [16, 34]. Other self-revealing techniques use the visualization of possible *gesture completion paths*: on-surface [4, 14] or mid-air [49].

The Charade and g-stalt systems combine natural and command set gestures. "Put-that-there" uses only one natural pointing gesture. Natural ergotic gestures learned while interacting with home appliances are utilized in [29, 41]. Natural as well as user-defined gestures are used in [50].

Therefore, we can differentiate the following taxonomy of gestures:

1. *Natural* (cultural, natural mapping)
2. *Command set*
3. *User-defined*
4. *Self-revealing*

Regarding the complexity of learning a gesture and self-revealing features, the concept proposed by Nancel et.al. [38] is of special interest; the authors called this *guidance through passive haptic feedback*. In their paper, different methods of input for mid-air interaction on wall-sized displays were compared. Mid-air and device based input were studied in particular. While executing the gesture, a user basically relies only on *proprioception*, the sense of the relative position of neighboring parts of the body and the ability to manipulate them. Device based input was

assessed with different types of devices, such as a mouse wheel or a touchpad. Contrary to the freehand techniques, device based techniques provide *passive haptic feedback* and give the user some limitations for the gesture execution. For example, when a finger operates the mouse wheel, there are only two directions to turn it, providing a kind of guidance for the movement. By using touch sensitive devices, the user is bound to a planar surface. Nancel et al. proposed to call this dimension a *degree of guidance* which can be considered contrary to the notion of degree of freedom. The authors showed that the degree of guidance strongly correlates with the interaction time.

There is a variety of prototypes available that enable mid-air haptic feedback, either based on magnetic field [54], ultrasound [26] or vibrators [15]. These methods were not investigated by Nancel et.al. although they would have had an impact on the results concerning the degree of guidance. Also, the state of the art of previously mentioned haptic feedback techniques is not enough to provide the same perceived feedback characteristic as presented when touching a real surface. In addition, these techniques are mostly obtrusive, limited in space, or they require wearables. Another technique for mid-air gesture guidance is based on the projection directly on the user's hand [49]. Apart from visual or haptic modalities for feedforward, auditory channel could be used to allow audio explorability of an interface. This can be well differentiated when comparing audio-explorability (feedforward) of pie menu based interfaces (e.g., [12]) and completion paths based interfaces (e.g. [4]). While radial menu afford the user an auditory exploration of the sectors by selecting them, state of the art completion paths interfaces can not support this feature. This is due to the fact that completion path interfaces are dynamically changing along with the performed gesture and therefore have no constant areas that can be explored.

Recovering from Errors

Giving the user a possibility to recover from an error is one of the main rules in HCI design. In the case of gesture interface the hand movement for undo should be memorable, concise, and unlikely to be performed incorrectly.

Fatigue

As a consequence of the fast and rare nature of microinteractions gestures, fatigue is not as much a problem in this kind of interfaces as it is in gesture interfaces for long term tasks. Issues concerning muscular tension discussed in gesture phrase correlates to some degree with the fatigue. The greatest tension occurs in activation and closure gestures. In addition, a gesture amplitude and thereby a number of muscles as well as the hand position play a crucial role for the fatigue issue. For instance, on-surface whole hand gestures would be less tiring than mid-air ones.

Unobtrusiveness

Ubiquitous computing implies the idea that technology is seamlessly integrated into the environment and is hidden from the user's perspective. Meanwhile, preserving original functionality of everyday objects used for the ubiquitous interaction as well as preserving their initial appearance are indispensable for well-designed smart environments.

Reliability

In our design principles we focus on real-world interfaces. Therefore, the overall robustness and usability of the system in a non-lab environment are of crucial importance. System developers should provide easiness of actuation, recalibration, and maintenance of the system.

Sustainability

Along with sustainability, commercial success of the interface is highly correlated with its energy efficiency. This implies an appropriate selection of the input method. Usually a sensing system based on the computer-vision demands a lot of CPU power.

EXEMPLARY APPLICATION OF THE DESIGN PRINCIPLES

Development of the exemplary system was done within the framework of the *Counter Entropy House*, a house built by students of different disciplines of *RWTH Aachen University*, for the *Solar Decathlon Europe 2012* Competition in Madrid. In this usage context, the three design principles for smart environments (*unobtrusiveness*, *reliability*, *sustainability*) have to be met by the physical setup. In order to give an overview of the system, we first describe the device setup and environment context of the interface.

Environment Context and Physical Setup

The developed system supports two main ways of interaction: performing microinteractions through a pie menu projected onto a kitchen table and quick selection of objects (light, windows) by pointing at them.

A depth-camera and a projector that enables the pie menu interaction are integrated into the lamp above the kitchen table. An additional depth-camera that traces pointing gestures is integrated into the wall facing a user sitting at the kitchen table. From this position the user has a good overview of the room and can select different objects like lights and windows. In addition, controllable objects were never placed in the one line-of-sight excluding spacial order (from the user), thereby allowing easy selection.

Sustainability

The use of depth-cameras and computer-vision for sensing contributed to a higher energy consumption, yet at the same time provided appropriate unobtrusive and ubiquitous characteristics of the interactive system. In order to compensate this suboptimal energy issue, a standby modus was proposed. The use of a PIR-sensor allows energy-efficient monitoring of the user's presence in the environment. After a certain inactive period, the depth-cameras as well as the corresponding computer vision software are suspended.

Interaction

The microinteractions concept is supported by the use of a self-revealing pie menu which appears around a user's hand as soon as an activation gesture is performed. This gesture consists of placing the hand on the surface and making a crumple-like gesture, shown in figure 3. Animation of sectors moving to the hand menu as well as an appropriate sound support the pulling metaphor of the gesture. Pre-studies revealed that this gesture has good characteristics concerning the robustness–learnability trade-off.

In the common pie menu concept the second hierarchy menu appears around the cursor as it touches the outer border of the sector in the top menu. In order to minimize the space requirements of an everyday surface (*unobtrusiveness*) nested menus appear in the same place as the top level. This feature is also necessary concerning the limited input/output area of the interactive system.



Figure 3: The camera-projector setup for on-surface gestures is seamlessly installed in the lamp. An activation gesture causes the appearance of a pie menu around the hand. Animation supports the metaphor of the activation gesture. The rotating feature allows to control the menu with the same gestures from all sides of the everyday surface.

The top level menu consists of multiple entries, see figure 4. An entry, e.g., “lights”, can be executed by dragging the fingertips throughout the corresponding sector, which causes the appearance of the submenu for “lights”. This contains entries such as “all on”, “all off” as well as entries to control lamps separately, each of which can be toggled by dragging the hand to the center of the menu. While two-level hierarchy of the pie menu allows fast navigation for novices, simple circular gestures are easy to remember for experts. In addition, bi-manual input is also supported by the system: on/off toggle entry for an object appears if a user points at it while the system is active.

Our *touch active zone* contributes to a simple *closure* by lifting the hand 10 cm above the surface. Our menu does not support incremental actions. Thereby, every microinteraction can be considered as a command shortcut whose execution terminates interaction with the menu. This fact and the use of the touch active zone allows us to limit complexity of the closure gesture.

Meanwhile, the pie menu is always rotated in relation to the forearm axis allowing to operate the menu with the same gestures from all sides of a surface. Furthermore, as soon as the available gestures have been learned by expert users, auditory feedback helps hearing the system's reactions without visual projection of the pie menu, thus allowing eyes-free interaction. The proposed rotation of the pie menu and its unobtrusive realization contribute to a universal installation of the system for multiple surface usage scenarios, especially for experts. The on-surface nature of gestures contributes to less *fatigue* and to a better *degree of guidance*.

Research in user experience over the last decade has shown, that along with traditional quality models non-utilitarian properties play also an important role in the acceptance and appealingness of the product [24]. Therefore, during tests we have also questioned users about perceived fun. User tests revealed that the demonstrated system for microinteractions has high usability and hedonic ratings [25], thus proving the helpfulness of the proposed design principles. The fact that the study was conducted in the field with visitors of Solar Decathlon during public tours determined the short duration of the experiment. Long-term experiments of the microinteraction system with real inhabitants could provide other results concerning hedonic ratings, as the perceived fun would change (decrease) over time.

SUMMARY

Our design principles, based on the fusion of different concepts from related literature and projects, might be a helpful instrument when developing new hand gesture interfaces for microinteractions. The concept of *gesture phrase* contributes to a more intuitive as well as more precise definition of interaction periods concerning hand tension and feedforward/feedback continuity.



Figure 4: Top level pie menu of the exemplary system for microinteractions.

The developed exemplary system and its overall evaluation success show the applicability of the proposed principles. Meanwhile, these principles are only a part of a more general and deeper design space whose exploration should be conducted further [9].

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